

Course Development Plan

Modernizing Course Structure, Materials, and Experimental Kit

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2103304 Automatic Control I (Fall 2026)

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Teaching Strategy

- Classical methods (Laplace, root locus, Bode) remain primary design tools
- Lectures stay broad and practical, with each topic linked to prior and upcoming weeks; heavy derivations move to handouts

Three coordinated changes

- **Modernized topics:** state-space form introduced in Week 2; threaded through analysis; focused design in Week 13 (pole placement, Ackermann)
- **Concrete physical anchoring:** every concept paired with a real-world example, **MATLAB live code**, and where possible a video or live demo
- **Two-half lecture rhythm:** theory in the first 1.5 hours; application, coding, and demos in the second

- **Continuity** from Linear Algebra and Differential Equations (2301216) — state space extends what students already know
- **Alignment to modern advanced topics:** data-driven, learning-based, MPC, and adaptive control are all state-space-native
- **Two-half lecture:** Consistent two-half lecture rhythm reduces cognitive load; modern examples, demos, lab connections, application engage students after the break

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Course Narrative — How the Weeks Connect

The throughline: performance specs $\rightarrow$ pole locations $\rightarrow$ controller design.

Three phases

- **Modeling (W1–3)** — represent the system: ODE, transfer function, state-space form.
- **Analysis (W4–7)** — understand behavior. **Week 4 defines where poles should be.**
- **Design (W8–13)** — multiple techniques for placing closed-loop poles at desired locations:
 - Root locus (W9–10): move poles by tuning gain
 - Frequency response (W11–12): margins, robustness
 - Pole placement (W13): **shortcut** from desired locations via state feedback

Weeks 1–7 (Pre-Midterm)

Wk	Theme	First half	Second half	Handout
1	Intro & Modeling primer	- Intro: motivation, open / closed, anatomy - m-s-d ODE; Laplace inverse	- LTI; convolution; impulse SS form preview	LA primer; Laplace tables
2	Modeling I (SS deep)	Cast m-s-d in SS; A, B, C, D e^{At}; eigenvalues	IVT / FVT; all 4 methods agree	Convolution; higher-order
3	Modeling II: TF $\leftrightarrow$ SS, MIMO	TF $\leftrightarrow$ SS; canonical forms	- MIMO; DC motor + gears - Linearization; equivalents	Koopman; PINNs
4	Time Response	1st / 2nd-order; poles / zeros	ζ, ω_n; rise / OS / settling From e^{At}	Dominant poles; non-min phase
5	Block Diagrams & Cascade	- Reduction; closed-loop TF SS feedback $A - BK$	Cascade control	Mason; networked
6	Stability + eigenvalues	- Routh–Hurwitz Shortcut to $\text{eig}(A_{cl})$	K_p range; $\text{eig}(A) = \text{poles}$	Routh edge cases; LMIs
7	Midterm Review (flipped)	Problems first	Synthesis second	Survival guide

Midterm covers Weeks 1–6.

Weeks 8–15 (Post-Midterm)

Wk	Theme	First half	Second half	Handout
8	SSE & PID intro	- SSE; system type - State aug. $\dot{x}_e = r - y$	Integral action; PID intro	Anti-windup; MRAC
9	Root Locus (RL) I	- The 8 rules Bridge: RL $\equiv$ SS char. poly	Sketching examples	RL multi-parameter
10	RL II: design	Lead / lag; PID via RL	Z–N; PZC pitfalls - Pole-placement preview	Z–N variants; MPC
11	Freq. Response I	Asymptotic Bode	- Bode plot construction SS frequency response	$\mathcal{H}_\infty$ preview
12	Freq. Response II	Nyquist; gain / phase margins	Bode-domain compensator design - Sensitivity S, T	System ID; $\mathcal{H}_\infty$
13	State Space	- State eqs (recap W2) - Controllability & observability	Pole placement; Ackermann - Luenberger observer	SS Advanced (~25pp)
14	Project Demo Day	Diagnostic-fix presentations	Tracking competition; judging	—
15	Final Review	Recap W8–13	Common pitfalls; exam strategy	Frontiers; “things forgotten”

Final exam: SSE · RL design · Bode · state space.

Acronyms Used in the Weekly Plan

Modeling & Analysis

- **ODE** — ordinary differential equation
- **LTI** — linear time-invariant
- **SS** — state space
- **TF** — transfer function
- **SS2TF / TF2SS** — state-space $\leftrightarrow$ transfer-function conversion
- **IVT / FVT** — initial / final value theorem
- **MIMO** — multi-input multi-output
- **DC** — direct current
- **m-s-d** — mass-spring-damper

Stability

- **LMI** — linear matrix inequality

Control Design

- **SSE** — steady-state error
- **PID** — proportional–integral–derivative
- **PI** — proportional–integral
- **RL** — root locus (*not* reinforcement learning, except W15)
- **Z–N** — Ziegler–Nichols
- **PZC** — pole-zero cancellation
- **OS** — overshoot

Modern / Advanced

- **MRAC** — model reference adaptive control
- **MPC** — model predictive control
- **LQR** — linear quadratic regulator
- **LQG** — linear quadratic Gaussian
- **LIP** — linear inverted pendulum
- **DCM** — divergent component of motion
- **PRBS** — pseudo-random binary sequence
- **PINN** — physics-informed neural network

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Per-Lecture Handouts

Goal: one self-contained handout per lecture so students can read, study, and review without a textbook.

Structure (10–20 pages depending on topic)

- Cover, learning objectives, suggested reading
- Story hook (same opener as lecture)
- Conceptual development — including content not covered entirely in class
- Worked examples on recurring demo systems
- Common misconceptions
- Practice problems with answers

Length scales with topic: short for simple weeks, longer for math-heavy lectures.

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Proposed (Fall 2026)

Homework	10%
Midterm Exam	30%
Final Project	20%
Final Exam	40%
Total	100%

Change from current:

- Midterm: 35% $\rightarrow$ 30%
- Final Project: 15% $\rightarrow$ 20%

The 5% shift reflects the larger role of the lab-based final project

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Servo Lab — Concept

Idea: a brushed DC motor coupled to a simple load (linear guide or pendulum), instrumented to (1) expose **electrical, thermal, mechanical, and sensor phenomena** for diagnosis, and (2) support **real-time controller design and tuning** (PID, lead/lag, state feedback) against the same plant.

Pedagogical Goal

Develop two coupled skills: *diagnosing servo misbehavior across physical domains* (e.g., when the motor oscillates, which domain is to blame?) and *designing a controller that meets target performance specs*.

- Mapped to course topics (sysID, P/PI, cascade, PID, state space).
- Reinforces the threaded state-space approach.
- Reusable across years and across courses.

Servo Lab — Status and Dual-Use

Status (June 2026): potential, not committed

Hardware design, BOM, and firmware are in progress. Whether the kit is ready for Fall 2026 depends on procurement timelines and prototype iteration.

Dual-use plan

The kit could be deployed in either:

- **2103304 Automatic Control I** — as an integrated lecture-aligned lab, or
- **2103360 ME Lab II** — as a standalone experiment module.

Backup: Weeks 1–7 (Pre-Midterm)

Acronyms: ODE = ordinary differential equation; LTI = linear time-invariant; SS = state space; IVT / FVT = initial / final value theorem; TF = transfer function; SS2TF (and TF2SS) = state-space $\leftrightarrow$ TF; MIMO = multi-input multi-output; DC = direct current.

Wk	Theme	First half	Second half
1	Intro + Modeling primer	Intro (~30 min): motivating examples; open vs closed loop; controller anatomy . Then mass-spring-damper ODE ; explicit + Laplace inverse	LTI systems (superposition, time-invariance); convolution integral; impulse response. <i>Modern add-on: brief touch on state-space (SS) form $\dot{x} = Ax + Bu$</i>
2	Modeling I (SS deep)	SS form, extensive: cast mass-spring-damper as $\mathbf{x} = [x, \dot{x}]^T$; build A, B, C, D ; e^{At} and eigenvalue responses	Initial / final value theorems (IVT / FVT); all four methods agree (explicit, convolution, Laplace inverse, IVT/FVT). <i>Modern add-on: same response from transfer function (TF) and e^{At} simulation</i>
3	Modeling II — TF $\leftrightarrow$ SS & MIMO	TF $\leftrightarrow$ SS equivalence (SS2TF, TF2SS) ; canonical forms; mech / elec first principles; through / across variables	Multi-input multi-output (MIMO) example; direct-current (DC) motor + gears; linearization; equivalent systems. <i>Modern add-on: controllable / observable canonical forms</i>
4	Time Response	1st / 2nd-order systems; poles / zeros. <i>Modern add-on: ζ, ω_n from eigenvalues of A</i>	ζ, ω_n ; rise / overshoot / settling. <i>Modern add-on: same response from TF and e^{At} simulation</i>
5	Block Diagrams & Cascade	Block reduction; closed-loop TF. <i>Modern add-on: SS feedback $A - BK$</i>	Cascade control ; feedback structure. <i>Modern add-on: cascade as nested SS loops</i>
6	Stability + eigenvalues	Routh–Hurwitz. <i>Modern add-on: Routh as a shortcut to check $\text{eig}(A_{cl})$ in LHP</i>	K_p range; eigenvalues introduced ($\text{eig}(A) = \text{poles}$). <i>Modern add-on: Lyapunov function preview</i>
7	Midterm (flipped)	Review Problems first	Synthesis second

Midterm covers Weeks 1–6 (modeling $\rightarrow$ stability).

Backup: Weeks 8–15 (Post-Midterm)

Acronyms (post-midterm): SSE = steady-state error; PID = proportional–integral–derivative; PI = proportional–integral; RL = root locus (not reinforcement learning, except in W15); Z–N = Ziegler–Nichols; PZC = pole-zero cancellation.

Wk	Theme	First half	Second half
8	SSE & PID intro	Steady-state error (SSE); system type. <i>Modern add-on: state augmentation $\dot{x}_e = r - y$</i>	Integral action; PID intro (proportional–integral–derivative). <i>Modern add-on: PI in SS form</i>
9	Root locus (RL) I	The eight rules. <i>Modern add-on: bridge identity $1 + KG(s) = 0 \iff \det(sI - (A - BK)) = 0$</i>	Sketching by hand examples
10	RL II — design	Lead / lag & PID via RL; generalized RL	Ziegler–Nichols (Z–N); pole-zero cancellation (PZC) pitfalls. <i>Modern add-on: pole placement as the shortcut (W13 preview)</i>
11	Frequency Response I	Asymptotic Bode; factor analysis	Building Bode plots from components. <i>Modern add-on: $G(j\omega) = C(j\omega I - A)^{-1}B + D$</i>
12	Frequency Response II	Nyquist; gain / phase margins	Non-min phase; Bode-domain compensator design (lead / lag tuning to hit a margin spec). <i>Modern add-on: sensitivity S and complementary sensitivity T</i>
13	State Space (SS)	State equations (recap from W2); controllability & observability	Pole placement; Ackermann's formula. <i>Modern add-on: Luenberger observer concept</i>
14	Project Demo Day	Diagnostic-fix milestone presentations	Tracking competition; judging. <i>Modern add-on: bonus credit for SS implementation</i>
15	Final Review	Recap Weeks 8–13	Common pitfalls; exam strategy. <i>Modern add-on: recap SS thread</i>

Final exam covers SSE, root-locus design, Bode design, and state space. Supplementary handout topics on next slide.

Backup: Supplementary Handout Topics (All Weeks)

Each lecture has a self-contained 10–20 page handout. Advanced material parked here so course depth is preserved without crowding the lecture.

Wk	Handout title	Content
1	Pre-course primer	Linear algebra refresher; course narrative one-pager; Laplace transform tables
2	Time-domain analysis	Convolution as analysis tool; higher-order systems; sine / exp response details
3	Modeling advanced	Advanced linearization; equivalent-system catalog; MIMO control preview; Koopman operator; PINNs for data-driven modeling
4	Pole-zero effects	Dominant poles; effect of zeros; non-minimum-phase preview; $\mathcal{H}_2$, $\mathcal{H}_\infty$ norms preview
5	Block algebra	Mason's gain formula; signal flow graphs; hierarchical & networked control preview
6	Stability tools	Routh edge cases; eigenvalue computation methods; Lyapunov direct method & LMIs preview
7	Midterm survival guide	Problem patterns across Weeks 1–6; common pitfalls
8	PID practical	Type-3+ systems; PID anti-windup; adaptive control preview (MRAC, gain scheduling)
9	RL extensions	Root locus with multi-parameter; advanced sketching cases
10	Root locus advanced	Generalized RL; PZC pitfalls deep dive; lead / lag design intuition; Z–N variants; MPC preview
11	Bode for hard cases	Bode for time delays and non-min phase; $\mathcal{H}_\infty$ as worst-case gain preview
12	FR practical	System ID primer (chirp / PRBS / step); sensitivity functions; $\mathcal{H}_\infty$ mixed-sensitivity
13	State space advanced (~25 pp)	Luenberger observer; LQR (linear quadratic regulator); humanoid LIP / DCM bridge; Kalman filter; LQG / $\mathcal{H}_\infty$; MPC
15	Exam prep	"Things students forget" one-pager; modern frontiers (reinforcement learning for control, safe learning, generative AI in control)

Weeks 14 (Demo Day) and Final Exam Week have no handout.

Backup: State-Space Approach

- 1 Introduced from Day 1** — SS form previewed in Week 1 alongside mass-spring-damper; deep dive in Week 2 (cast m-s-d as $\mathbf{x} = [x, \dot{x}]^\top$; build A, B, C, D ; e^{At} and eigenvalue responses).
- 2 Threaded as a basic modern lens (W3–W12)** — TF $\leftrightarrow$ SS equivalence (W3); ζ, ω_n from $\text{eig}(A)$ (W4); SS feedback $A - BK$ in block diagrams (W5); $\text{eig}(A)$ = poles with stability (W6); $G(j\omega) = C(j\omega I - A)^{-1}B + D$ in frequency response (W11).
- 3 Classical methods drive design (Weeks 8–12)** — root locus and Bode remain the primary design tools.
- 4 One focused state-space week (Week 13)** — state equations, controllability / observability, pole placement, Ackermann's formula. Pole placement is the **shortcut**: once desired pole locations are known (from performance specs in Week 4), Ackermann delivers the feedback gain directly.
- 5 Advanced topics in supplementary handouts** — LQR, Luenberger observers, humanoid LIP / DCM bridge, Kalman filter, LQG / $\mathcal{H}_\infty$, MPC.

Backup: Textbook Selection

Text	Best for	Weakest at	Role
Nise	Undergrad pedagogy; root locus; frequency response; case studies	State space (Ch. 12) lighter than Ogata	Primary
Ogata	State space; MATLAB; mathematical rigor	Dense for first encounter	Secondary (L13)
Franklin, Powell, and Emami-Naeini (FPE)	Design-first framing; best digital control; balanced classical/modern	Senior-level; more demanding for first encounter	Strong optional all-rounder
Dorf & Bishop	Design emphasis; historical context; case studies	Busy layout	Tertiary (L10/L12)
Phillips & Parr	Digital / sampled-data; mathematical cleanliness	Weakest pedagogical scaffolding	Not adopted

Recommendation: Nise primary (matches existing decks); Ogata as L13 reference; Dorf for design vignettes. FPE as optional all-rounder or consolidating replacement for Ogata + Dorf.

Backup: Reference Curricula

- **Nise; Dorf–Bishop** (classical-first): state space in final chapters. Current 2103304 follows this pattern.
- **Åström–Murray, Feedback Systems** (Caltech): state space in Chapters 2–3; both threads alive throughout.
- **Hespanha, Linear Systems Theory**: state space as the spine; transfer functions derived from it.

Proposed for 2103304

Closer to Nise structure than Åström–Murray, with explicit threading. **A conservative middle path.**